

Family Solanaceae

Habit and Vegetative Features

Mostly herbs and shrubs; stem herbaceous, sometimes tuberous (potato); leaves alternate, simple, exstipulate, reticulate venation.

Inflorescence and Flower

Solitary or cymose inflorescence; flowers bisexual, actinomorphic.

Floral Whorls

Calyx: 5 united sepals, valvate, persistent. Corolla: 5 united petals, valvate. Androecium: 5 epipetalous stamens. Gynoecium: bicarpellary, syncarpous, superior, axile placentation.

Fruit and Economic Importance

Fruit is a berry or capsule with endospermous seeds. Economic uses: food (Tomato, Brinjal, Potato), spice (Chilli), medicine (Belladonna, Ashwagandha), fumigatory (Tobacco), ornamental (Petunia).

Diagnostic Vegetative Features

Solanaceae members are typically identified by their alternate simple leaves, exstipulate condition, and often hairy stems, alongside characteristic solitary or cymose inflorescences arising in leaf axils.

Distinguishing Solanaceae from Related Families

Unlike Fabaceae (with vexillary aestivation and free carpels), Solanaceae shows valvate aestivation and a syncarpous bicarpellary gynoecium, making floral dissection key to accurate family identification.

Solanaceae Floral Formula Breakdown

Whorl	Composition
Calyx	K(5)
Corolla	C(5)
Androecium	A5, epipetalous
Gynoecium	G(2), syncarpous, superior

Economic Importance of Solanaceae

- Food: Tomato, Brinjal, Potato
- Spice: Chilli
- Medicinal: Belladonna, Ashwagandha
- Fumigatory: Tobacco
- Ornamental: Petunia